Question Paper

Exam Date & Time: 29-Nov-2025 (09:30 AM - 12:30 PM)

MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025
SUBJECT: CSS 2104/CSS_2104 DIGITAL SYSTEMS COMPUTER ORGANIZATION
(SPL: COMPUTER SCIENCE AND ENGINEERING / ARTIFICIAL INTELLIGENCE / CYBER SECURITY / DATA SCIENCE / INFORMATION TECHNOLOGY)

Marks: 50 **Duration: 180 mins.**

Answer all the questions.

1A) Analyze the architectural differences between RISC and CISC processors. Write a CISC-style program to compute the dot product of two vectors given by (4)

$$\sum_{i=0}^{n-1} A(i) \times B(i)$$

Course Outcome: "CO5" / Cognitive Level: "Analysis"

1B) Simplify the following function and implement it using only NOR gates only. (3)

$$F(A,B,C,D) = \pi\, M(0,3,5,6,8,10) + D(2,13,14)$$

Course Outcome: "CO1" / Cognitive Level: "Application"

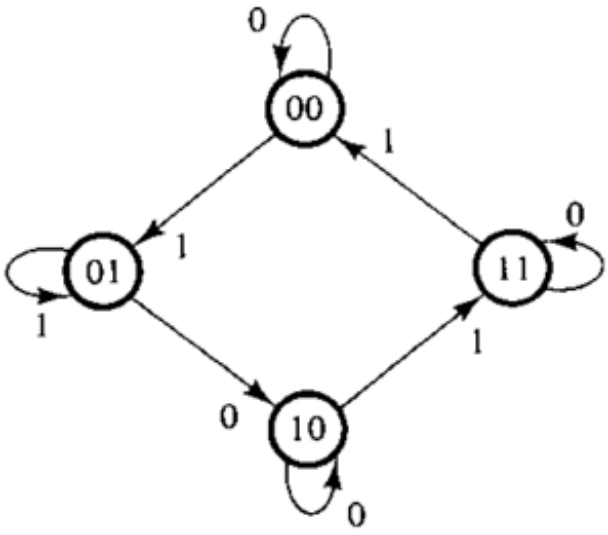
1C) Design a 3-bit magnitude comparator for two unsigned binary numbers (3)

$$A = a_2 a_1 a_0 \text{ and } B = b_2 b_1 b_0.$$

Derive the final Boolean expressions and draw the equivalent logic circuit diagram

Course Outcome: "CO1" / Cognitive Level: "Create"

2A) Analyze the following state diagram and design the corresponding synchronous sequential circuit using JK flip-flops. (5)



Course Outcome: "CO2" / Cognitive Level: "Create"

2B) Illustrate a 4-bit asynchronous ripple counter with a neat diagram using JK Flip flops and explain the working of the same. (3)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

2C) Write behavioral Verilog code that represents an eight-bit ring counter. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

3A) The parameters of the computer memory system are specified as follows: (4)
Main memory size = 8K blocks, Cache memory size = 512 blocks, Block size = 8 words. Determine the size of tag field of the main memory address under the following conditions.
i) Fully associative mapping
ii) Direct mapping
iii) Set associative mapping with 16 blocks/set

Course Outcome: "CO4" / Cognitive Level: "Analysis"

3B) Explain the concept of memory addresses in an Instruction Set Architecture. Also, discuss the difference between byte addressing and word addressing. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

3C) Perform 13 X -6 using Booth's algorithm. Write all the necessary steps. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

4A) What is pipeline? Demonstrate 5 stage organization of pipelining with Block diagram. (4)

Course Outcome: "CO4" / Cognitive Level: "Application"

4B) Perform the following in IEEE 32-bit format. Show the result in the IEEE 32 bit format. (3)
A = 86.625 and B = 10.125.
Find A + B.

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

4C) Analyze the purpose of decoder in Wilke's design of microprogrammed control unit. (3)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

5A) Write the register transfer description algorithm for 4x4 booth multiplication. Draw the state diagram and indicate all the necessary control signals. Also draw the logic diagram of the Booths multiplier controller using sequence controller, decoder and counter. (5)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

5B) Design the processing section for the register transfer description given below by establishing the necessary control points. (3)

```
Declare registers A[4], B[4], D[4];
Declare bus outbus [4];

Start:       A←0, B←0, D←4;
Loop:     if A < B and D[0] = = 0   then go to transferA
             go to Decrement;
transferA:  A ← B;
            B ← A + B;
Decrement: D ← D - 1;
             if D < > 0 then go to loop
            outbus =A;
Halt:       go to Halt
```

Course Outcome: "CO3" / Cognitive Level: "Create"

5C) For the following boolean expression, draw the PLA. (2)
i) F=ABC′+ABC+A′BC+AB′C+AB′C′
ii) F1=A′B+BC′ and F2=AB′+A′C

Course Outcome: "CO3" / Cognitive Level: "Application"

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Event Name	Subject
END TERM EXAM-ODD SEMESTER 2025 MIT MANIPAL	CSS 2104:DIGITAL SYSTEMS AND COMPUTER ORGANIZATION

#	Course Outcome	Description	Marks
1	1	CO 1	6
2	2	CO 2	10
3	3	CO 3	19
4	4	CO 4	8
5	5	CO 5	7
		Total	**50**

* NA - CO Not Selected

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END TERM EXAM-ODD SEMESTER 2025 MIT MANIPAL	CSS 2104:DIGITAL SYSTEMS AND COMPUTER ORGANIZATION

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	4	11(22%)

4	Analysis	7	25(50%)
5	Create	1	3(6%)
6	Evaluate	3	11(22%)
	Total	15	50 (100%)